heat energy & Temperature

* heat energy [Q] ⟹ is the energy transfered
or Thermal energy
↓
joul

from body to another due to Temp. difference

* Temperature [T] ⟹ physical quantity That express hot & cold.

Celsius — C°
Kelvin — K [International unit]
Fehranheit — °F

Note: ⟹ * if a body gain heat energy (Q) Temp. will Rais up.

* if a body loss heat energy (Q) Temp. will go down.

Q↑ Temp↑ v↑ K.E↑

Q↓ Temp↓

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Temp. units
- → Celsius [°C]
- → Kelvin [K]
- → Fehrenheit [°F]

① $T_K = T_C + 273$

② $T_F = \frac{9}{5} T_C + 32$

③ $\Delta T_K = \Delta T_C = \frac{5}{9} \Delta T_F$

	Boiling point of water	freezing point of water	ΔT
°C	100 °C	0 °C	100
°K	373 °K	273 °K	100
°F	212 °F	32 °F	180

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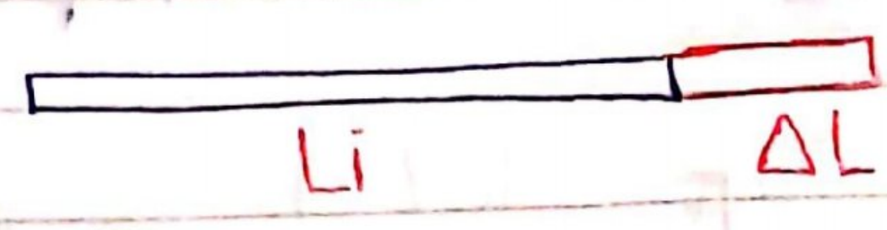
Thermal expansion

1] Linear Thermal expansion

$$L_F = L_i + \Delta L$$



$$\Delta L = \alpha L_i (T_F - T_i)$$

$L_i \rightarrow$ intial Length
$L_F \rightarrow$ Final length
$T_i \rightarrow$ intial Temp.
$T_F \rightarrow$ Final Temp.
$\alpha \rightarrow$ Linear expansion Cellicient
$\Delta L \rightarrow$ Change exp. in length

3] Volume Thermal expansion

$$\Delta V = 3\alpha V_i (T_F - T_i)$$

$$\boxed{B = 3\alpha}$$

$B \rightarrow$ volume expansion Cell.

$V_i \rightarrow$ intial Volume
$\Delta V \rightarrow$ Change exp. in Volume

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Sheet [1]

Lec

1] A nurse measure the temperature of a patient to be 107 °F

a] What is this temperature on the celsius scale?

$T_F = 107\,°F$, $T_C = ??$, $T_K = ??$

$$T_F = \frac{9}{5} T_C + 32$$

$$107 = \frac{9}{5} T_C + 32 \Rightarrow T_C = 41\,°C$$

$$T_K = \underset{41}{T_C} + 273 \Rightarrow T_K = 314\,°K$$

Lec

2] The Temperature difference between the inside and the outside of a home on a cold winter day is 57.0 °F. Express this difference on a] the celsius scale and b] the kelvin scale

$\Delta T_F = 57\,°F$

$$\Delta T_C = \Delta T_K = \frac{5}{9} \cdot \Delta T_F = \frac{5}{9} \cdot 57 = 31.66$$

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4] The boiling point of liquid hydrogen is 20.3 K at atmospheric pressure. what is this temperature on a] the celsius scale and b] the Fahrenheit scale?

$T_K = 20.3\,°K$, $T_K = T_C + 273$

a]. $T_C = T_K - 273 = 20.3 - 273 = -252.7°C$

b] $T_F = \frac{9}{5} T_C + 32 = \left[\frac{9}{5} * -252.7\right] + 32$

$= -422.86\,°F$

3] Convert the following temperatures to their values on the Fahrenheit and Kelvin scales:

a] the sublimation point of dry ice, $-78.5°C$

b] human body temperature, $37°C$.

a] $T_C = -78.5°C$

$T_F = \frac{9}{5} T_C + 32 = \frac{9}{5}[-78.5] + 32 = -109°F$

$T_K = T_C + 273 = -78.5 + 273 = 195\,K$

b] $T_F = \frac{9}{5} T_C + 32 = \left(\frac{9}{5} * 37\right) + 32 = 98.6°F$

$T = T_C + 27.15 = (37 + 273) = 310\,K$

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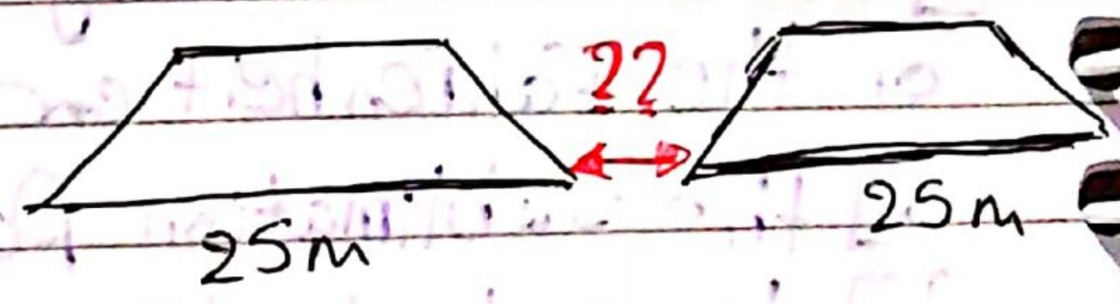
5] The Concrete sections of a certain superhighway are designed to have a length of 25 m. The sections are poured and cured at 10 °C. What minimum spacing should the engineer leave between the sections to eliminate buckling if the concrete is to reach a temperature of 50 °C ?

$L_i = 25m$, $T_i = 10°C$

$\alpha = 12 \times 10^{-6}$, $T_F = 50°C$, $\Delta L = ??$

$$\Delta L = \alpha L_i (T_f - T_i)$$

$$= 12 \times 10^{-6} \times 25 \times (50 - 10) = 0.012m$$



6] A copper telephone wire has essentially no sag between poles 35 m apart on a winter day when the temperature is -20 °C How much longer is the wire on a summer day when the temperature is 35 °C ?

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$L_i = 35\,m$, $T_i = -20\,°C$, $T_f = 35\,°C$

$\alpha = 16.5 \times 10^{-6}$, $L_F = ??$

$\Delta L = \alpha L_i (T_f - T_i)$

$= (16.5 \times 10^{-6})(35)(35+20)$

$= 0.0317\,m$

$L_F = L_i + \Delta L$

$L_F = 35 + 0.0317 = 35.031\,m$

7] A pair of eyeglass frames is made of epoxy plastic. At room temp. 20°C, the frames have circular lens holes 2.2 cm in radius. To what temperature must the frames be heated if lenses 2.21 cm in radius are to be inserted in them? The average coefficient of linear expansion for epoxy is $1.3 \times 10^{-4}\,°C^{-1}$.

$L_i = 2.2\,cm$ $T_i = 20\,°C$

$L_f = 2.21\,cm$ $T_F = ??$

$\alpha = 1.3 \times 10^{-4}\,°C^{-1}$, $\boxed{\Delta L = L_F - L_i = 0.01}$

$\Delta L = \alpha L_i (T_f - T_i)$

$0.01 = (1.3 \times 10^{-4})(2.2)(T_F - 20)$

$T_F = 54.9\,°C$

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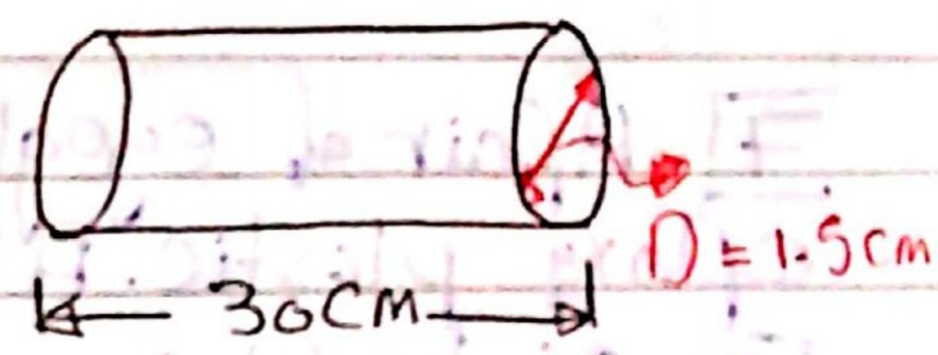
9] The active element of a certain laser is made of a glass rod 30cm long and 1.5cm in diameter. Assume the average coefficient of linear expansion of the glass is equal to $9 * 10^{-6}\ {}^{\circ}C^{-1}$. If the temperature of the rod increases by 65°C, what is the increase in [a] its length [b] its diameter, and [c] its volume.

diameter = 1.5cm

$L_i = 30cm$

$\alpha = 9 * 10^{-6}\ {}^{\circ}C^{-1}$

$\Delta T = 65\ {}^{\circ}C$



[a] $\Delta L = \alpha L_i \Delta T = 9 * 10^{-6} * 30 * 10^{-2} * 65$
$= 0.1755 * 10^{-3}\ m$

[b] $\Delta D = \alpha D_i \Delta T = 9 * 10^{-6} * 1.5 * 10^{-2} * 65$
$= 8.77 * 10^{-6}\ m$

[c] $\Delta V = ?$

V_i = length * base Area
$= (30 * 10^{-2}) * \pi (0.75 * 10^{-2})^2 =$
$53 * 10^{-6}\ m^3$

$\Delta V = \underbrace{\boxed{3\alpha}}_{\beta} V_i \Delta T = (3 * 9 * 10^{-6}) * (53 * 10^{-6}) * 65$
$= 6.093 * 10^{-6}\ m^3$

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The 70-liter (L) steel gas tank of a car is filled to the top with gasoline at 20°C. The car sits in the Sun and the tank reaches a temperature of 40 °C. How much gasoline do you expect to overflow from the tank?

(Coefficient of Volume Expansion for gasoline, $\beta = 950 \times 10^{-6}\ (C°)^{-1}$)
{Coefficient of Linear Expansion for steel, $\alpha = 12 \times 10^{-6}\ (C°)^{-1}$ }

Q(2)

(A) for steel tank :

$$\Delta V = \beta V_i \Delta T = (3 * 12 * 10^{-6})(70)(20)$$
$$= 0.0504\ L$$

for gasoline :

$$\Delta V = \beta V_i \cdot \Delta T = (950 * 10^{-6})(70)(20) = 1.332\ L$$

gasoline over flow $= 1.33 - 0.0504 = 1.2796\ L$

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